MIL-H-45586A

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SUPERSEDING
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(See Section 6)

MILITARY SPECIFICATION

HOWITZER, LIGHT, TOWED, 105MM: M102

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of 105MM light towed Howitzer, M102, consisting of the M137A1 cannon, M37A1 Recoil Mechanism, and the M31 carriage.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-P-98 - Paint; Stencil, Flat.

Military

MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.
MIL-H-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Testing.
MIL-H-6088 - Heat Treatment of Aluminum Alloys.
MIL-I-6866 - Inspection, Penetrant, Method of.
MIL-S-6872 - Soldering Process, General Specification for.
MIL-H-6875 - Heat Treatment of Steels (Aircraft Practice, Process for).

FSC 1015

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MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series: General Specification for.
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel Packaging and Packing of.
MIL-E-16053	-	Electrodes, Welding, Bare, Aluminum Alloys.
MIL-M-43719	-	Marking Materials and Markers, Adhesive, Elastomeric, Pigmented General Specification for.
MIL-W-45205	-	Welding, Gas Metal - Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor.
MIL-C-45579	-	Cannon, 105MM Howitzer: M137A1.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Desposition of

STANDARDS

Federal

FED-STD-595	-	Colors.

Military

MIL-STD-9	-	Screw Thread Conventions and Methods of Specifying.
MIL-STD-930	-	Identification Marking of U. S. Military Property.
MIL-STD-171	-	Finishing of Metal and Wood Surfaces.

DRAWINGS

Rock Island Arsenal

F8428300	-	Howitzer, Light, Towed, 105MM, M102.
F12000725	-	Recoil Mechanism, 105MM, Howitzer, M37A1
F8433200	-	Carriage Assembly, 105MM Howitzer, M31.

PUBLICATIONS

Rock Island Arsenal

P8428300	-	Packaging Data Sheet, for Howitzer, Light, Towed, 105MM, M102, W/E.
IEL8428300	-	Inspection Equipment List Howitzer, Light, Towed, 105MM, M102

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standard Institute
ANSI B46.1 - Surface Texture
ANSI Y14.5 - Dimensioning and Tolerancing for Engineering Drawings

(Application for copies should be addressed to The American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N. Y. 10017).

American Welding Society
AWS A2.0 - Welding Symbols

(Application for copies should be addressed to the American Welding Society, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The howitzer shall conform to the materials, construction and design requirements specified herein, and on Drawing F8428300.

3.2.1 Heat treatment. Unless otherwise specified on the applicable drawing, heat treatment for aluminum alloys shall conform to MIL-H-6088 and for steel shall conform to MIL-H-6875.

3.2.2 Dimensions. Dimensions and tolerances of parts and assemblies of the recoil mechanism and carriage shall conform to the detailed drawings of Drawing F8428300. Interpretation of the dimensions and tolerances shall be in accordance with ANSI - Y14.5.

3.2.3 Surface texture. Surface texture shall be as specified on the applicable drawings. Interpretation of surface roughness symbols shall be as defined in ANSI B46.1.

3.2.4 Protective finishes.

3.2.4.1 Inorganic finishes. Parts requiring an inorganic finish shall be finished or coated as indicated on the applicable drawings and in accordance with MIL-STD-171.

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3.2.4.2 Painting. The parts and assemblies shall be painted as specified on the drawing and in accordance with MIL-STD-171. The visible surfaces of attaching hardware (nuts, screws, washers, etc.), exposed unpainted surfaces of mounting faces, or blemished areas shall be primed and painted to match surrounding surfaces after final assembly. To avoid necessity of painting exposed surfaces of cadmium plated hardware (including screws, nuts, and washers), the hardware may be dyed with a supplementary chromate finish, Olive Drab 24087 of Federal Standard 595 (see 6.3.2).

3.2.5 Welding. Interpretation of weld symbols specified on the applicable drawings shall be as defined in AWS A2.0. Actual production of weldments shall not proceed prior to Government approval of the contractor's welding procedures and test specimens of the applicable welding specifications. All plug welds shall be flush and free of snags or protrusions. Grinding flush is not required unless specified on the drawing or in the welding specification. Weld spatter and similar excess weld shall be removed.

3.2.5.1 Welding, aluminum alloy. Welding of aluminum alloy parts shall conform to MIL-W-45205. Type 5356 welding electrode of MIL-E-16053 shall be used for aluminum alloy welds. Type 5183 or 5556 welding electrodes of MIL-E-16053 may be used if Type 5356 welding electrode is unavailable.

3.2.5.2 Welding, ferrous metals. Welding of steel parts and assemblies shall be in accordance with the specification referenced on the applicable drawings.

3.2.5.3 Pre-set data (Box Trail Weldment). The pre-set data for stress relieving of the Box Trail weldment before machining, shall be in accordnace with the applicable drawing.

3.2.5.4 Magnetic particle inspection. When magnetic particle inspection is indicated on the drawing, it shall be in accordnace with MIL-I-6868.

3.2.5.5 Penetrant inspection. When penetrant inspection is indicated on the drawing, it shall be in accordance with MIL-I-6866.

3.2.6 Soldering. Soldering of parts shall be as indicated on the applicable drawing and in accordance with MIL-S-6872.

3.2.7 Threads. Threads shall conform to the thread series and class indicated on the applicable drawing, and be in accordance with MIL-STD-9. Unless otherwise specified, coated or plated threads shall conform to the coated thread data specified in MIL-S-7742.

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3.2.7.1 Torque, threaded fasteners. Unless otherwise specified on the applicable drawing, threaded fasteners shall be torqued in accordance with the following torque requirements:

Thread Size	Torque
No. 4 - 48-UNF	6-8 lb. foot
No. 6 - 32-UNC	7-9 lb. foot
No. 8 - 32-UNC	14-17 lb. foot
No. 10 - 24-UNC	20-25 lb. foot
No. 10 - 32-UNF	20-25 lb. foot
1/4 - 20-UNC	70-80 lb. foot
3/8 - 16-UNC	20-27 lb. foot
1/2 - 13-UNC	50-70 lb. foot
5/8 - 11-UNC	90-120 lb. foot

3.3 Performance and product characteristics.

3.3.1 Recoil mechanism. The completed recoil mechanism shall conform to the characteristics specified on Drawing F12000725 and the requirements of this specification. The recoil mechanism shall be filled with hydraulic fluid, bled, and refilled until all air and measurable moisture are eliminated. The initial and subsequent filling of the hydraulic system shall be with hydraulic oil conforming to MIL-H-5606 or MIL-H-6083 and filtered through a 5 micron filter. Prior to and after gymnastication, the hydraulic system shall be free of all magnetic or abrasive particles that may cause damage to the inner surfaces of the operating mechanism, or malfunction of the system. Testing shall be as specified in 4.5.3.2.

3.3.1.1 Mechanical gymnastication. The recoil mechanism shall be mechanically gymnasticated 25 cycles in recoil and counter recoil. Recoiling parts shall be moved 49 inches plus or minus 1 inch and return to battery position. The recoil mechanism shall function without erratic movement or malfunction of the system. Seepage of oil through the mating surfaces shall not exceed two (2) cubic centimeters in any 24 hour period.

3.3.1.2 Friction. Friction of the recoil rod when assembled in the Recoil Cylinder, and the floating pistion assembly, shall be within the limits specified in Table I after assembly, after gymnastication, and after proof-firing.

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TABLE I

Friction in Pounds Per Square Inch
Hydraulic Pressure

Part Name	Maximum Limits
Rod, 8432833	180 P.S.I.
Floating piston assembly	90 P.S.I.

3.3.1.3 Nitrogen leakage. The nitrogen filling end of the pressurized recuperator cylinder shall show no evidence of nitrogen leakage when observed for 5 minutes.

3.3.1.4 Reserve oil indicator. The reserve oil indicator piston stem shall function without evidence of interference throughout its entire range of travel. There shall be no evidence of oil leakage. Oil wetting around the piston stem is permissable.

3.3.1.5 Foamed oil and contaminated oil. After exercising the recoil mechanism 25 cycles by mechanical gymnastication, or after proof-firing, oil withdrawn slowly from the recoil mechanism into a straightsided glass beaker shall be in accordance with the following: Within the first minute after withdrawel, the oil shall be five percent or less foam by volume; when exposed to the atmosphere for 15 minutes, the aeration bubbles shall disappear; after exposure to the atmosphere for one hour, there shall be no sediment at the bottom of the glass beaker. Testing shall be as specified in 4.5.3.2 and 4.5.3.5.

3.3.1.6 Indicator and control assembly. The torque required to rotate the Indicator and Control Assembly (F8432840) thru 360 degrees shall not exceed 20 pound-foot after proof-firing.

3.3.2 Carriage. The completed carriage shall conform to the characteristics specified on Drawing F8433200, and be in accordance with this specification. The recoil mechanism M37A1 and cannon M137A1, with a dummy round of ammunition in the chamber, or their combined equivalent weight, shall be assembled to the carriage to determine the requirements of 3.3.2.1 thru 3.3.2.8, with the carriage emplaced on a level surface. Testing shall be as specified in 4.5.3.3.

3.3.2.1 Elevating mechanism. The elevating mechanism shall elevate the cannon tube of the carriage from a position of minus five degrees, to plus seventy-five degrees. The torque required to elevate or depress the howitzer at any point throughout the operating range shall not exceed 75 pound-inches at the handwheel shaft. The maximum backlash of the elevating handwheel shall not exceed 45 degrees of rotation at any position in either direction of elevation or depression.

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3.3.2.2 Traversing mechanism. The traversing mechanism shall be capable of traversing the carriage, in either direction a full 360 degrees. The torque required to traverse the carriage in either direction shall not exceed 75 pound-inches on the handwheel shaft. The maximum backlash of the traversing handwheel shall not exceed 30 degrees of rotation in either direction.

3.3.2.3 Variable recoil actuating mechanism. The variable recoil actuating mechanism shall be assembled and adjusted in accordance with F8433200. When elevating and depressing the cannon, the mechanism shall move freely without bind or looseness. Adjustments of the linkage shall be such that the indicator line on the end of the regulator lies within the 0 degree to 10 degree range (from vertical) markings on the end of the recuperator cylinder when the cannon is at 0 degree elevation and between the 30 degree to 40 degree (from vertical) markings when the cannon is at a minimum of 40 degrees elevation.

3.3.2.4 Handbrakes. The handbrakes, when engaged, shall prevent wheel rotation while the carriage is in ascending or descending travel position on a 15 degree slope, the surface of which must be such that the tires will not slip. Travel position is defined as with the traversing wheel tire not touching the surface. When the handbrake is disengaged, the wheels shall turn freely without drag. To provide proper surface contact of the brake lining, a maximum of 1/32 inch may be removed from the surface thickness of the lining.

3.3.2.5 Lunette assembly. The lunette assembly shall be capable of being secured in the firing position (up) or the towing position (down), by the quick-release pin shown on the applicable drawing. The lock plate incorporated in the lunette assembly shall permit the lunette to be fixed or free, depending on conditions.

3.3.2.6 Firing platform. The firing platform shall be capable of allowing the carriage to rotate a full 360 degrees in traverse. The lock assembly shall secure the platform to the box trail assembly pivot, and shall be secured in its engaged position by the lock pin attached to the platform.

3.3.2.7 Wheel actuator. The wheel actuator shall be capable of lowering the carriage to firing position and raising the carriage to travel position. The actuator shall be self-locking throughout the range of operation and free of noise and chatter. The maximum torque required at the input shaft to raise and lower the carriage shall not exceed 320 inch-pounds.

3.3.2.8 Locking devices. All locking and fastening devices for securing component parts shall be positive in action.

3.3.3 Fire control.

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3.3.3.1 Fire control instruments. With the elevating trunnions of the carriage cross leveled by having the cannon tube tracking a plumb line within 0.1 mil, to a minimum elevation of 50°, the fire control instruments shall meet the requirements of 3.3.3.2 thru 3.3.3.6. Testing shall be as specified in 4.5.3.4.

3.3.3.2 Alignment of telescope mount. Elevate the cannon tube from 0 to 1330 mils in 100 mil steps. Re-level the M134 Mount in both directions for each 100 mil step and check for deviation of line sight from the aiming point. Maximum deviation from 0 to 900 mil elevation is 0.5 mil and from 900 to 1330 mils a total of 1.0 mil.

3.3.3.3 Telescope mount rigidity. The total movement of the vertical cross hair of the panoramic telescope relative to the aiming point shall not be more than 0.3 mil for zero cant and zero elevation and 1.75 mils for zero cant and 1100 mils elevation after a 20 lb. tangential load is applied to each side of the telescope support bracket as shown in Figure 3.

3.3.3.4 Alignment of fire control quadrant. The cross level vial bubble of the fire control quadrant shall remain centered within the width of ± 1/2 vial graduation space when the weapon is activated to center the elevation level vial bubble in conformance with wettings of the elevation counter.

3.3.3.5 Accuracy of elevation counter. Prior to proof-firing, the elevation counter readings of the fire control quadrant, with respect to the center line of the cannon tube, shall not exceed 0.5 mils excluding backlash at any counter setting. After proof-firing the carriage, the allowable deviation shall not exceed 1.0 mil, excluding backlash.

3.3.3.6 Backlash. Backlash shall not exceed 0.7 mils at any elevation counter reading of the fire control quadrant.

3.3.4 Proof-firing.

3.3.4.1 Recoil mechanism. The recoil mechanism shall be subjected to proof-firing by firing four (4) Government standard M1 Cartridge, 105mm (33 pound inert projectile), or equivalent, in accordance with Table II. The recoil mechanism shall function without evidence of malfunction or failure. Testing shall be as specified in 4.5.3.5.1.

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Table II - Proof-Firing

Number of Rounds	Chamber Pressure Actual PSI	Percent Charge (Reference)	Simulated Elevation Degree	Recoil Length Inches	Cycle Time, Total Seconds: 80°F and over Min	80°F and over Max	79°F and under Min	79°F and under Max
1	39,000 ± 1,000	100	0°	43-50.75	1.0	3.0	1.0	4.0
1	44,800 ± 1,000	115	0°	43-50.75	1.0	3.0	1.0	4.0
*1	44,800 ± 1,000	115	75°	26-31.50	1.0	3.0	1.0	4.0
1	39,000 ± 1,000	100	0°	43-50.75	1.0	3.0	1.0	4.0

*Note: This round may be fired at 0 degrees elevation with the variable recoil arm locked for short recoil.

3.3.4.2 Carriage. The carriage shall be subjected to proof-firing by firing three(3) rounds of Government standard M1 Cartridge, 105MM (33 pound inert projectile) or equivalent, in accordance with Table III. The carriage shall function without evidence of malfunction or failure. Variable in trammel markings from their initial position after proof-firing. shall not be more than 1/32 inch. Testing shall be as specified in 4.5.3.5.2.

Table III - Proof-firing

No. of Rounds	Chamber Pressure Actual P.S.I.	Percent Charge (Reference)	Elevations Degrees
1	39,000 ± 1,000	100%	0°
*1	44,800 ± 1,000	115%	75°
1	39,000 ± 1,000	100%	0°

*Note: This round may be fired at 0 degrees elevation with the variable recoil arm locked for short recoil.

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3.3.5 Road test. The completely assembled carriage shall be capable of withstanding the following 50 mile road test without damage, permanent deformation or loosening of parts. The trammel marking requirement used for proof-firing (see 3.3.4.2), shall also apply for road testing. Testing shall be as specified in 4.5.3.6.

35 miles at 35 miles per hour (MPH), over hard surface roads.
10 miles at 15 miles per hour, over dirt roads.
5 miles at 10 miles per hour, over unimproved or cross country roads.

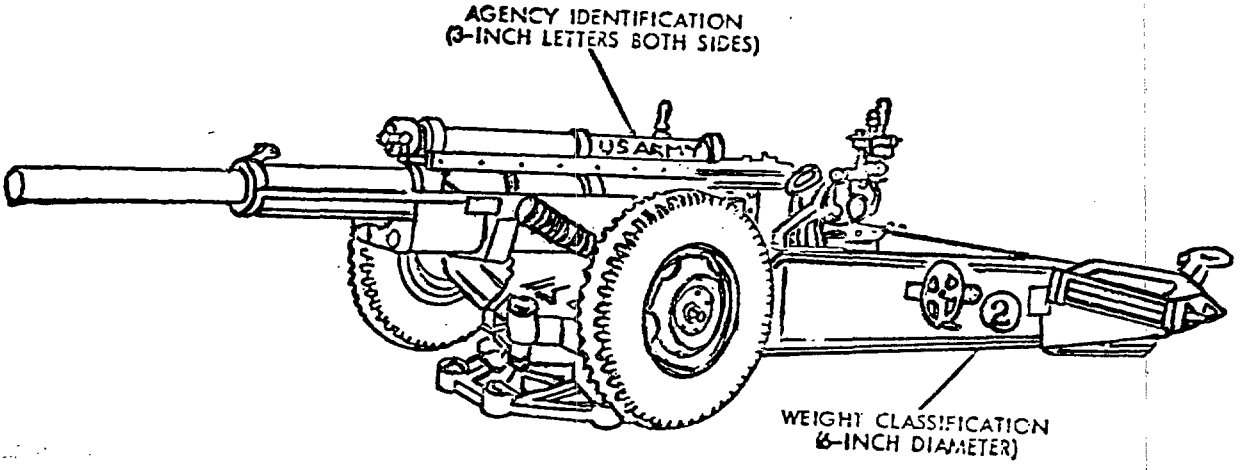
3.3.6 Boresight retention. After the proof-firing and road tests, the line of sight of the panoramic telescope shall maintain its original setting in azimuth with respect to the weapon tube within 0.3 mil. The direct fire telescope shall maintain its original setting with respect to the weapon tube within 0.8 mil in azimuth and 0.8 mil in elevation. Testing shall be as specified in 4.5.3.7.

3.4 Cannon. Cannon, 105MM Howitzer, M137A1 and parts thereof shall conform to MIL-C-45579.

3.5 Identification and instructional marking. Unless otherwise specified (see 6.2), agency identification marking, tire pressure marking and weight classification marking shall be applied in accordance with Figure I. Other marking, including application of part numbers and instructional marking, shall be as specified on the applicable drawings and in accordance with MIL-STD-130. Serial numbers shall be furnished by the procuring activity, (see 6.2).

3.6 Workmanship. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations and missing operations which may affect serviceability, functioning, operation, appearance or safety. Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding) or other similar practices shall not be permitted without prior approval of the contracting officer.

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Agency identification

Color: Lusterless white #37855
Location: As illustrated

Weight classification

Number: 2
Color: Lusterless black number (2) #37033 on a six-inch diameter background of lusterless yellow #33538
Location: Left side of the box trail just to the rear of the traversing handwheel

Tire pressure

Color: Lusterless white #37875
Location: Any visible portion of the wheel support arm
Letters and numerals: The marking will consist of numerals representing the proper tire pressure preceded by the letters TP
Size: Letters and numerals will be one inch

General data

Markings shall be applied using stencil paint in accordance with TT-P-98 or pressure sensitive adhesive materials in accordance with MIL-M-43719.

All letters and numerals shall be of the gothic style and of the maximum practical size unless otherwise specified.

Colors shall be as specified in accordance with FED-STD-595 (see 6.3)

Figure 1. Size and location of markings for 105-mm light towed howitzer M102

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspections as necessary to determine that all the requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified herein, the formation, size and presentation of inspection lots of howitzers and parts shall be in accordance with MIL-STD-105. Howitzers shall be assembled from lots of component parts that have met all inspection requirements specified herein.

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of component parts and concurrent repair parts shall be performed in accordance with the criteria specified in the contract (6.2). The contractor's examination of these parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

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4.5.2.2 Howitzers. Visually and manually examine each howitzer to determine conformance with the applicable requirements. Each step in the examination shall include a visual examination for proper cleaning, the presence of the specified protective coating, and to determine the general quality, completeness of manufacture, assembly, and workmanship. The examination provisions shall be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Reinspections of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling, and that adequate inspection records are maintained. Howitzers failing to meet the requirements shall be rejected.

4.5.2.2.1 Parts and assemblies. Parts and assemblies shall be visually and manually examined to determine compliance with the materials, construction and design requirements of 3.2 and the applicable requirements of 3.2.1 through 3.2.7 using the following examination and test specified in 4.5.2.2.1.1 through 4.5.2.2.1.8.

4.5.2.2.1.1 Heat treatment. Examination and tests of heat treated components shall be in accordance with MIL-H-6088 or MIL-H-6875, as applicable.

4.5.2.2.1.2 Welds and welding. Examination and tests of welded parts and welding procedure for aluminum alloy shall be in accordance with MIL-W-45205. Examination and tests of welding procedures for steel parts shall be in accordance with the specification referenced on the applicable drawings.

4.5.2.2.1.3 Magnetic particle inspection test. When magnetic particle inspection is specified on the drawing, the test shall be in accordance with MIL-I-6868.

4.5.2.2.1.4 Penetrant inspection test. When penetrant inspection is specified on the drawing, the test shall be in accordance with MIL-I-6866.

4.5.2.2.1.5 Protective finishes. Examination of painted surfaces and inorganic finishes shall be in accordance with the specifications specified on the applicable drawings and MIL-STD-171.

4.5.2.2.1.6 Soldering. Examination and test of soldered parts shall be as specified in MIL-S-6872.

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4.5.2.2.1.7 Visually examine all markings required by the final assembly drawing and Figure 1 to assure they are clear and legible.

4.5.2.2.1.8 Visually examine the identification plate for compliance with the applicable drawing, for secure assembly (see 3.5), serialization, and to assure that all pertinent data has been applied.

4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete howitzer. If test requirements cited herein are not met, acceptance of the howitzer shall be deferred and the contractor shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine howitzers, partially assembled howitzers, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned howitzers for retest.

4.5.3.2 Recoil mechanism testing. Each completely assembled recoil mechanism shall be tested for performance and product characteristics (see 3.3.1 and 3.3.1.1 thru 3.3.1.6) using the test methods specified in 4.6.1. Failure of the recoil mechanism to meet the requirements shall be cause for rejection.

4.5.3.3 Carriage testing. Each completely assembled carriage shall be tested for performance and product characteristics (see 3.3.2 and 3.3.2.1 thru 3.3.2.8) using the test methods specified in 4.6.2. Failure of the carriage to meet the requirements shall be cause for rejection.

4.5.3.4 Fire control testing. Each completely assembled carriage shall be tested for fire control (see 3.3.3.1 thru 3.3.3.6) using the test methods specified in 4.6.3. Failure of the fire control to meet the requirements shall be cause for rejection.

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4.5.3.5 Proof-firing.

4.5.3.5.1 Recoil mechanism testing. Each recoil mechanism shall be proof-fired (see 3.3.4.1) using the test method specified in 4.6.4. Proof-firing for acceptance of the recoil mechanism may be combined with that of the carriage (see 4.6.4). After completion of the proof-firing test, the recoil mechanism shall be subjected to the test methods of 4.6.1.2 thru 4.6.1.5. The indicator and control assembly shall meet the torque requirements to determine compliance with 3.3.1.7. Failure of the recoil mechanism to meet the requirements shall be cause for rejection.

4.5.3.5.2 Carriage testing. Each carriage shall be proof-fired (see 3.3.4.2) using the test method specified in 4.6.4. Proof-firing of the recoil mechanism may be combined with this test. After completion of the proof-firing test, the recoil mechanism shall be subjected to the test methods of 4.6.1.2 thru 4.6.1.5. The indicator and control assembly shall meet the torque requirements to determine compliance with 3.3.1.7. Failure of the recoil mechanism and carriage to meet the requirements shall be cause for rejection.

4.5.3.6 Road testing. One complete carriage from the first five (5) produced and one from each 50 thereafter shall be subjected to the road test (see 3.3.5) using the test method specified in 4.6.5. Failure of the carriage to meet the requirements shall be cause for rejection.

4.5.3.7 Boresight retention. Each carriage shall be tested for the boresight retention (see 3.3.6) using the test method specified in 4.6.6 after completion of the proof-firing and road test specified in 4.6.4 and 4.6.5 respectively. Failure of the carriage to maintain the boresight retention after proof-firing and the road test shall be cause for rejection.

4.5.3.8 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.2). This will include chemical analysis and physical tests of materials, tests of protective finish, heat treatment, bonding, and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.4 Packaging examination and testing. Unless otherwise specified (see 6.2). The packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.5 Inspection equipment.

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4.5.5.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.2), responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment, IEL 8428300, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.5.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

4.6.1 Recoil mechanism test. Each recoil mechanism shall be checked for nitrogen pressure, recoil oil, and other operating parameters specified on Drawing F12000725 prior to subjecting the recoil mechanism to the mechanical gymnastication test.

4.6.1.1 Gymnastication test. Each recoil mechanism shall be mechanically gymnasticated as specified in paragraph 3.3.1.1. Upon completion of gymnastication, the recoil mechanism shall be subjected to testing as specified below:

4.6.1.2 Friction test. Manometer readings shall be taken and recorded on the recoil rod assembly and the floating piston assembly to determine that friction of these components does not exceed the maximum limits specified in 3.3.1.2 (see 6.4).

4.6.1.3 Nitrogen leakage test. A solution of soap and water shall be applied to the nitrogen filling end of the pressurized recuperator cylinder. There shall be no evidence of soap bubbles when the cylinder is observed for a period of five (5) minutes (see 3.3.1.3).

4.6.1.4 Reserve oil indicator test. Prior to replacement of the reserve oil, the reserve oil indicator shall be checked for freedom of movement throughout its entire range of travel. Upon replenishing the reserve oil, the reserve oil indicator shall be visually checked for conformance to 3.3.1.4.

4.6.1.5 Foamed oil test. Immediately after mechanical gymnasticating, and after proof firing, oil shall be withdrawn slowly from the recoil mechanism into a glass beaker. The recoil mechanism shall be rejected, if after exposure to the atmosphere for one (1) minute the quantity of foam exceeds five (5) percent by volume, and when aeration bubbles in the oil do not disappear after exposure to the atmosphere for 15 minutes, also, the oil withdrawn shall be inspected for contaminated oil. Contaminated oil is when sediment appears at the bottom of the glass beaker after exposure to the atmosphere for one (1) hour (see 3.3.1.5).

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4.6.2 Carriage. The completely assembled carriage being tested, shall consist of the recoil mechanism (M37A1) and the cannon tube (M137A1) with a dummy round of ammunition (approximate weight 38 pounds) inserted in the cannon tube chamber. The carriage shall be emplaced on a level surface.

4.6.2.1 Elevating mechanism test. With the dummy round inserted in the chamber of the cannon, a torque wrench graduated in pound-inches shall be utilized for measuring loads at the shaft of the handwheel to determine that the elevating handwheel loads do not exceed the torque value specified in 3.3.2.1. Measurements shall be taken at the following elevations, both in elevating and depressing: -5°, +0°, +5°, +10°, +15°, +25°, +32°, +34°, +38°, +45°, +55°, +65°, and +75°. The methods for taking measurements is as follows: Depress weapon to -5 degrees and take the torque reading on the nut both clockwise and counterclockwise. Then elevate the weapon to the next station, zero degrees, and repeat taking torque readings on the nut both clockwise and counterclockwise. Repeat this operation at all stations while elevating the weapon between -5 degrees and maximum elevation. The same method is to be used at all stations while depressing the weapon from maximum elevation to -5 degrees. The handwheel shall not exceed the backlash specified in 3.3.2.1.

4.6.2.2 Traversing mechanism test. With the dummy round inserted in the chamber of the cannon, a torque wrench graduated in pound-inches, shall be utilized for measuring at the shaft of the handwheel to determine that the traversing handwheel loads do not exceed the torque values specified in 3.3.2.2. Traversing measurements shall be taken at 90 degree intervals while traversing the howitzer with the tube at zero degrees elevation. The handwheel shall not exceed the backlash specified in 3.3.2.2.

4.6.2.3 Variable recoil mechanism test. With the dummy round inserted in the chamber of the cannon, check the installation and adjustment of the variable recoil linkage in accordance with the requirements of 3.3.2.3. Depress the gun tube to 0 degrees elevation. The indicator line on the end of the regulator rod must be within the white 0 degree to 10 degree zone on the end of the recuperator cylinder. Elevate the gun tube to 40 degrees or greater. The indicator line must be within the white 30 degrees to 40 degrees zone on the end of the recuperator cylinder. Depress the gun tube to zero degrees elevation. The indicator line on the end of the regulator rod must return to the 0 degree to 10 degree zone. The variable recoil mechanism shall move freely without bind or excessive looseness throughout the entire range of travel.

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4.6.2.4 Handbrakes test. The handbrakes of the completed carriage shall be tested for conformance with 3.3.2.4. The requirement may also be checked by using an 800 ± 80 pound push or pull on the lunette with the weapon traveling position on a level surface, or checked by applying a minumum of 500 ± 50 pound-foot torque to each wheel.

4.6.2.5 Firing platform test. The firing platform shall be secured in its locked position on the carriage. The completed carriage shall be lowered onto its firing platform and the carriage shall be rotated 360 degrees.

4.6.2.6 Wheel actuation test. The completed carriage shall be subjected to emplacement of the assembled howitzer on its firing platform. The following criteria shall be established before starting the test. The cannon tube with assembled breech shall be in battery position; the parking brakes shall be disengaged; and the wheel locking pins removed. A torque measuring device shall be used to measure the torque required to lower the weapon to firing position, and raise the weapon to travel position. During the cranking operation of lowering and raising the weapon, the operator shall stop cranking for a period of one minute ± 10 seconds at any three given points (two points while the firing platform is off the ground and one point when the firing platform is on the ground) in the lowering and in the raising operation, to determine self-locking characteristics of the actuator (see 3.3.2.7).

4.6.3 Fire control tests.

4.6.3.1 Fire control instruments. The following fire control test equipment shall be used.

a. Mechanical jacks for leveling the carriage trunions.
b. A plumb line approximately 1/16 inch in diameter shall be suspended less than 12 inches from the end of the cannon tube. The line shall be shielded from wind currents and the plumb bob shall be immersed in a container of oil to act as a dampener.
c. A boresight target shall be constructed on non-warping and non-shrinking material (Figure 2). Place the target 50 feet ± 6 inches from the fire control instruments in a plane perpendicular to the cannon bore at zero weapon elevation. The reference line on the target shall be plumb. The height of the target above ground shall be such that at zero elevation, the centerline of the cannon bore shall fall on the cannon aiming point within plus or minus 1/8 inch.

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d. An approved borescope shall be used. This shall be a telescope with a cross hair reticle mounted in a self aligning adapter to be inserted in the breech of the cannon so that the crosshairs of the telescope shall have a separate focus for reticle and field and be adjustable from 5 to 5,000 yards. The reticle of the telescope shall be collimated with respect to the outside diameter of the adapter within 0.1 mil. The telescope shall have a magnification of 4X or higher.
e. Parallax shields (peep sight) with a centrally located 1/16 inch maximum diameter hole, shall be placed over the eyepiece of the panoramic telescope and elbow telescope to eliminate parallax.

4.6.3.2 Fire control, checks and adjustments.

a. The carriage trunnion shall be cross-leveled so that the cannon tube will track the plumb line within 0.1 mil through the full elevation. This shall be done by using mechanical jacks and the borescope (see 4.6.3.1(d)) inserted in the breech and setting the muzzle cross hairs (.03 ± .01 dia) on the center of the plumb line at zero weapon elevation. The cannon tube should be elevated to approximate maximum elevation but not less than 50° when checking plumb travel.
b. Level the first three calibers of the cannon tube using cannon tube leveling fixture P/N 8776834 and an inspection aid (calibrated gunner's quadrant M1A1).
c. Place a target (Figure 2) 50 feet ± 6 inches from the fire control instruments. Sight through the borescope and align the target lines A and B in coincidence with the borescope reticle cross lines. Ascertain that the telescope mount and fire control quadrant level vial bubbles are centered. Sight through the panoramic telescope and align the reticle to its respective target lines C and D by means of the respective knobs. Set the sliding reticle of the elbow telescope to 0 and align the reticle to its respective target lines E and F making all the adjustments with the boresight adjusting screws of the quadrant. Use of parallax shield for each optical system is recommended to eliminate parallax.

4.6.3.3 Alignment of telescope mount. Alignment of the panoramic telescope mount is determined by sighting through the panoramic telescope while elevating the weapon in 100 mil steps and re-leveling the M134 Mount in both directions for each 100 mil step. The intersecting point of the reticule shall not deviate from the intersecting point on the target by more than 0.5 mils from 0 - 900 mils elevation or a total of 1.0 mils from 900 - 1330 mils elevation. The following procedure shall be followed when checking for deviation of line of sight from the aiming point.

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a. Pitch adjust M134 Mount. When centering bubble on pitch level vial, move the pitch knob so that the bubble moves from right to left when the observer faces the vial.
b. Crosslevel adjustments M134 Mount. When centering the crosslevel vial, move the crosslevel knob so that the bubble moves from left to right when the observer faces the vial.
c. All panoramic telescope readings shall be taken in the following manner:

(1) The vertical crosshair shall always be moved to the left of the aiming point a minimum of 10 mils and then moved to the right until the vertical cross hair falls on the aiming point. Note if the vertical cross hair over runs the aiming point to the right, the vertical cross hair must again be moved to the left and be repositioned on the aiming point prior to reading the amount of deviation from the aiming point.
(2) All deviations from the aiming point must be read on the upper counter of the panoramic telescope.

4.6.3.4 Telescope mount rigidity. Apply a tangential load of 20 pounds at the right side of the telescope support bracket, perpendicular to the axis of the telescope (see Figure 3) and release gradually to avoid disturbing the setting. Place the vertical cross hair of the panoramic telescope in coincidence with the target point G, Figure 2. Apply the same load tangentially at the left side of the bracket and gradually release. The total movement of telescope vertical cross hair shall not exceed the requirements of paragraph 3.3.3.1.2.

4.6.3.5 Alignment of the fire control quadrant and accuracy of elevation counter.

a. With the carriage oriented in accordance with 4.6.3.2 and with an inspection aid on the quadrant pads, elevate the cannon tube to any elevation setting and measure the elevation with the inspection aid. Turn the elevating mechanism of the quadrant to level the elevation level vial. (Do not override this level position.)
b. When the level vial bubble indicates level, the counter shall produce a reading in coincidence with the reading obtained on the inspection aid within the limit specified in 3.3.3.1.4, and the cross level vial movements shall be within the limits specified in 3.3.3.1.3. Repeat this procedure at a minimum of eight elevation settings through the full excursion of elevation travel.

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4.6.3.6 Backlash test. This test will be performed by elevating the weapon and M14 quadrant to maximum elevation by means of the elevating mechanism of the weapon. Elevate the gun tube to the maximum elevation reading taken during the test for alignment of fire control quadrant (4.6.3.5). Exercise care not to override this position. Rotate the elevation knob of the quadrant to indicate a level reading of the elevation level vial. The reading produced on the counter must coincide with the elevation setting of the gun tube and also coincide with those elevation readings taken during the test for alignment of fire control quadrant within the limits specified in 3.3.3.4.

4.6.4 Proof-firing test. Proof-firing of carriage shall consist of firing both standard rounds and specific pressure rounds of ammunition. Proof-firing shall be performed by the Government (see 6.2). The carriage shall be proof-fired in accordance with Table III of 3.3.4.2, or in accordance with Table II of 3.3.4.1 in event proof-firing is accomplished simultaneously with the recoil mechanism. Prior to proof-firing, trammel point markings as specified in 3.3.4.2, shall be established on the unit under test. Variance in trammel markings from their initial position after the proof-firing shall not be more than 1/32 inch. Visual examination for cracks, weld failures, deformation and other forms of damage shall be conducted on each unit (in conjunction with trammel markings) at the conclusion of the firing program. The proof-firing and trammel point markings prior to and after proof firing shall be recorded (see 6.3.3).

4.6.5 Road test. The completed carriage (less dummy round) shall be coupled to a vehicle equipped with standard pintle to determine compliance with the requirements specified in 3.3.5. Prior to road testing, trammel marks shall be established on the unit under test. Visual examination for cracks, weld failures, deformations and other forms of damage shall be conducted. Variance in trammel markings from their initial position after the road test shall not be more than 1/32 inch.

4.6.6 Boresight retention test. All lines of sight of the optical instruments, including borescope, shall be secured in coincidence with their respective aiming points as defined in 4.6.3.2. The exact position of the carriage shall be marked to insure repositioning after performing the road test and proof-firing test to determine possible displacement of any lines of sight. Upon completion of the proof-firing test (see 4.6.4) and the road test (see 4.6.5), the carriage shall be positioned over the original marked location and the reticle of the borescope shall be placed in coincidence with its respective target aiming point (lines A and B of Figure 2). Observe the respective lines of sight to determine their relationship to the original settings. The panoramic telescope and direct fire telescope lines of sight shall not have deviated more than the tolerance specified in 3.3.6.

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4.6.7 Cannon. Examination and tests of the cannon, 105MM Howitzer, M137A1 and parts thereof shall be as specified in MIL-C-45579.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. A pilot pack shall consist of a complete packaged in accordance with Packaging Data Sheet P8426974 to the level of protection specified in the contract (see 6.2), packed level C and forwarded in accordance with 3.1.

5.2 Preservation, packaging, packing and marking. Shall be preserved, packaged, packed and marked in accordance with Packaging Data Sheet P8426974 for the level of protection specified in the contract (see 6.2).

5.3 Repair parts. Repair parts shall be preserved, packaged, packed and marked in accordance with the Packaging Data Sheet or other requirements for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The M102 Howitzer is a lightweight, towed, 105MM Field Artillery Weapon suited for close support operations. This weapon has a low silhouette, is highly mobile, and is capable of full 360 degree traverse, with an elevation range of minus 5 degrees, to plus 75 degrees. This weapon is also capable of being air-lifted and transportable by helicopter.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. List of drawings and specifications pertinent to the howitzer, showing applicable revision dates.
c. Inspection criteria for components (see 4.5.2.1 and 4.5.3.8).
d. Shipping instructions for first article and pilot pack (see 3.1 and 5.1).
e. List of acceptance inspection equipment to be furnished the contractor (see 4.5.5.1) and responsibilities for other Government property to be furnished the contractor.
f. Packaging examination and testing, if different (see 4.5.4).
g. Shipping instructions for proof-firing (see 4.6.4).
h. Marking if different, and block of serial numbers(see 3.5).
i. Selection of applicable levels of preservation, packaging, packing and marking (see 5.1 and 5.2).

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j. Packaging of repair parts (see 5.3).
k. Disposition of Government furnished property.

6.3 Color. The following color chips of FED-STD-595 for use as a color guide for painting (see 3.2.4.2 and Figure 1) may be obtained from:

General Services Administration
Business Service Center
Region 3
Seventh and D Streets, S.W.
Washington, D.C.

No. 24087 Olive Drab, Semi-Gloss
No. 35538 Yellow-Lusterless
No. 37038 Black-Lusterless
No. 37855 White-Lusterless

6.4 Contract data requirements. When warranted, reports of the results of the recorded readings obtained in 4.5.1.1.(a), 4.5.1.2(a) and 4.5.2.9 shall be specified for delivery on the DD Form 1423 included in the contract.

6.5 Supersession data. This specification included the requirements of Purchase Description RIAPD-697 dated 26 June 1967.

6.6 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.7 Unless otherwise specified (see 6.2), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

Custodians:
Army - WC
Navy - MC

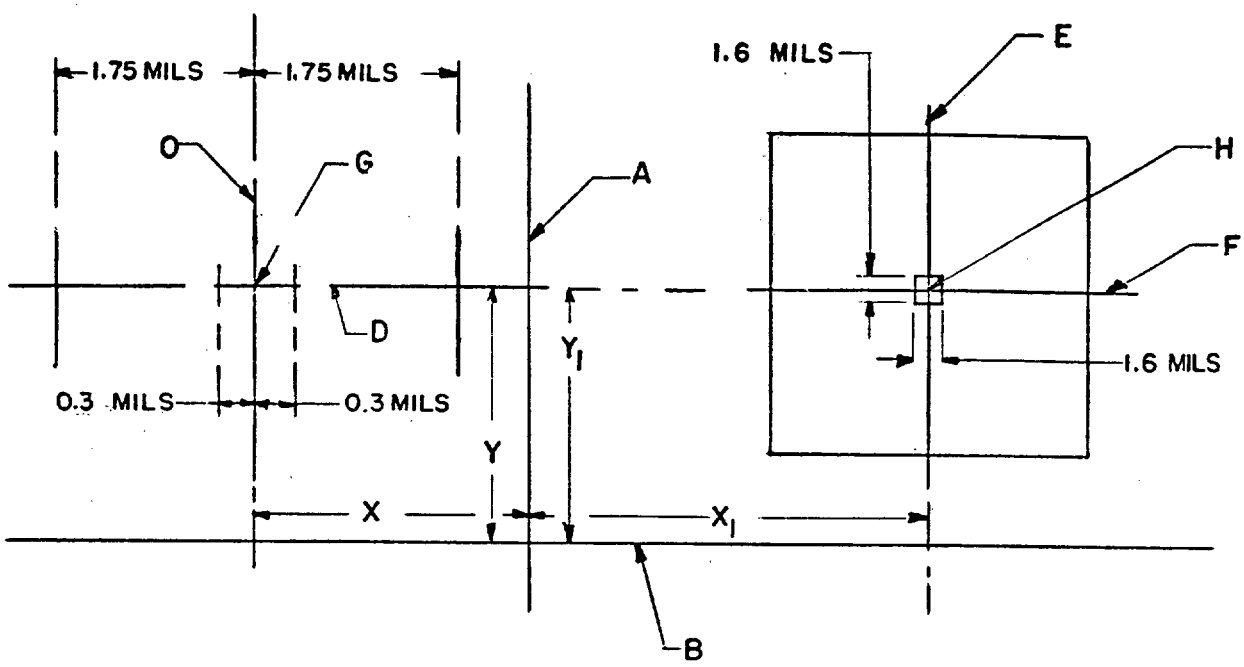
Preparing Activity:
Army - WC

Project Number 1015-0262

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PANORAMIC TELESCOPE: M113

ELBOW TELESCOPE: M114



A & B - Centerline of Gun Bore

G - Optical axis of Pan. Tel: M113

Y - Vertical Displacement between Pan. Tel. (Optical Axis) & ℄ of Gun Bore

X - Horizontal Displacement between Pan. Tel. (Optical Axis) & ℄ of Gun Bore

H - Optical Axis of Elbow Tel: M114

Y_1 - Vertical Displacement between Elbow Tel. (Optical Axis) & ℄ of Gun Bore

X_1 - Horizontal Displacement between Elbow Tel. (Optical Axis) & ℄ of the Gun Bore

Figure 2

Target for fire control alignment w/respect to gun bore

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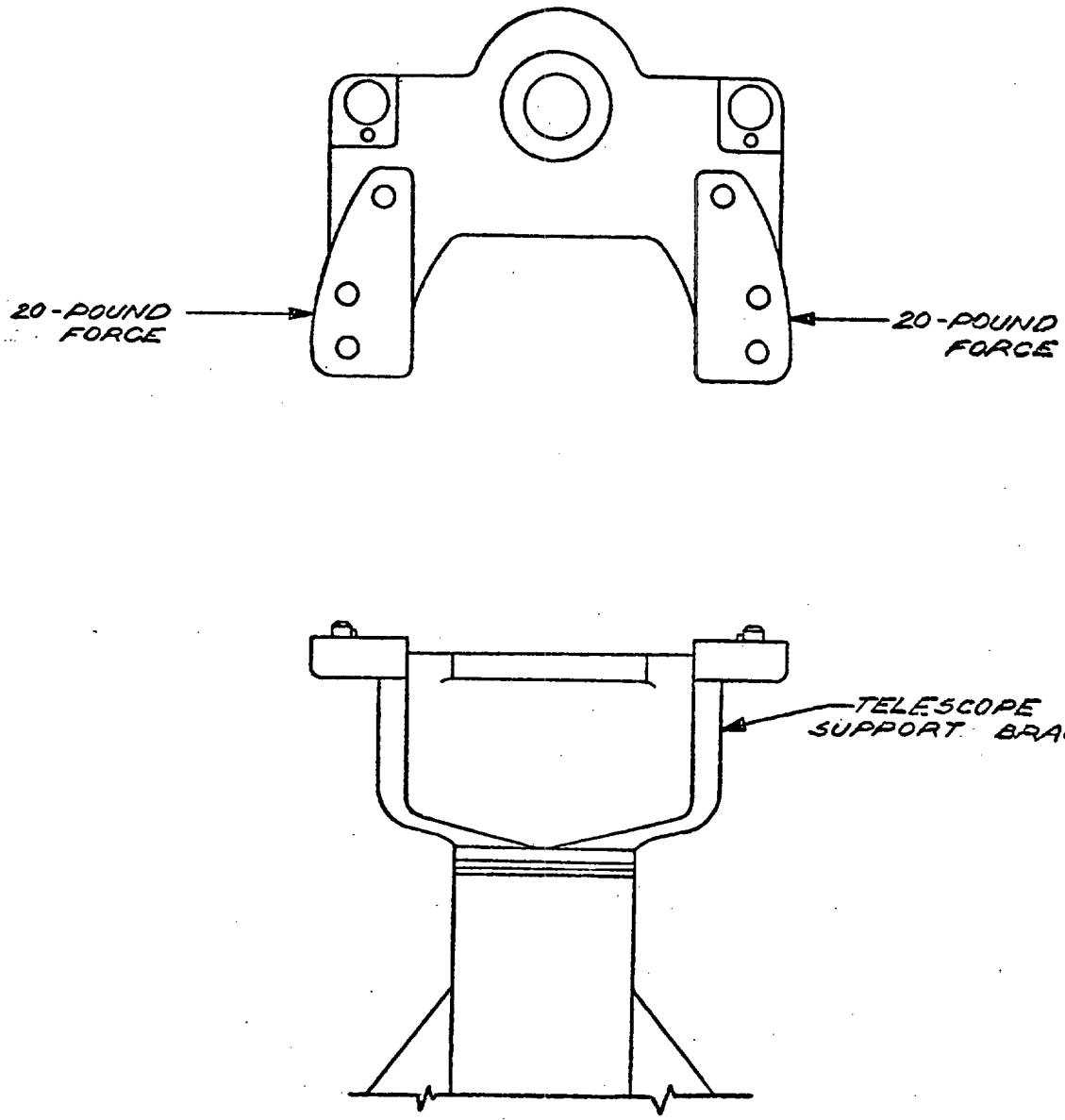


FIGURE 3

APPLICATION OF TANGENTIAL LOAD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-H-45586A; Howitzer, Light, Towed, 105MM: M102

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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